

- (b) (i) Use Fig. 4.1 and Fig. 4.2 to determine the general expression for the internal energy U of the gas when it has pressure p and volume V .

$$U = \dots\dots\dots [1]$$

- (ii) An ideal gas at thermodynamic temperature T contains N molecules.

Use your answer in (b)(i) and the equation of state for an ideal gas to deduce an expression for U in terms of N and T . Identify any other symbols you use.

$$U = \dots\dots\dots [2]$$

- (c) Determine expressions, in terms of X and Y , for the work W done on the gas during:

- (i) change AB

$$W = \dots\dots\dots [1]$$

- (ii) change CD.

$$W = \dots\dots\dots [1]$$

- (d) Use your answers in (c) and the first law of thermodynamics to determine an expression, in terms of X and Y , for the net thermal energy Q supplied to the gas during one full cycle ABCDA. Explain your reasoning.

$$Q = \dots\dots\dots [3]$$

[Total: 10]

